

A Four-Sector TESS Ephemeris for TOI-3505.01 and a Reconstruction of a 2022 Ground-Based Null Observation

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Context. Transit ephemerides lose predictive precision when their period or epoch is superseded, which can move a short ground-based observing sequence away from the event it was intended to cover. TOI-3505.01 (TIC 390988385) remains listed as a planet candidate.

Aims. We reanalyzed a 2022 George Mason University (GMU) 0.8 m R-band sequence, combined the four public TESS sectors, tested the stability of an updated linear ephemeris, and investigated why the ground sequence did not show the cataloged signal.

Methods. We performed differential aperture photometry on 263 quality-reviewed ground measurements, tested the scheduled window, and scanned the whole sequence while holding the duration at a published TESS value and floating the midpoint. We injected the corresponding published depth at every admissible midpoint to evaluate phase sensitivity in the observed residuals. We measured individual TESS events with a local linear baseline and an exposure-integrated trapezoid. The primary timing fit used one light curve per event: QLP for Sectors 14 and 41 and two-minute SPOC photometry for Sectors 54 and 81. We tested sector aggregation, event and sector deletion, selection thresholds, a quadratic timing model, and one public MuSCAT2 timing as an external control. We also propagated public 2021 timing markers with the superseded period recorded in a later ExoFOP report.

Results. The fixed 2022 schedule window gives a depth of -0.658 ± 0.395 ppt and therefore no transit-like dimming. The whole-sequence scan contains no fitted event above 3 σ under the formal weighted-least-squares covariance. Exact-depth injections exceed that formal threshold at 80.6% of admissible catalog-duration midpoints and 95.2% of SPOC-duration midpoints, so the sequence does not support exclusion at every phase. The adopted ephemeris uses 27 TESS mid-transit times over 5.05 yr and gives $P = 2.9151516 \pm 0.0000049$ d and $T_0 = 2459793.53115 \pm 0.00088$ BJD_{TDB}. The delete-one-sector sensitivity scale is 0.0000073 d, and a quadratic term is not favored ($\Delta\text{BIC} = +1.71$). The adopted ephemeris places the relevant event 20.31 h before the 2022 schedule midpoint. In contrast, propagating the public 2021 ingress, midpoint, and egress by 96 cycles at the superseded 2.9174250 d period reproduces all three schedule markers within 56.1 s.

Conclusions. The 2022 null observation is consistent with a stale schedule window rather than evidence against the recurring TESS signal. The updated timing solution is robust to the tested controls, but the unresolved 0.517 arcsec companion and uncertain dilution prevent a physical radius measurement. This work does not validate or confirm TOI-3505.01.

Keywords: planetary systems · eclipses · ephemerides · photometry · methods: data analysis · stars: individual (TOI-3505)

Status boundary. The public snapshot used here lists both the TESS and TFOP Working Group dispositions as PC and PC, respectively. Throughout this paper, “candidate” is intentional; none of the analyses below constitutes statistical validation or confirmation.

1. Introduction

The Transiting Exoplanet Survey Satellite (TESS) searches bright stars for periodic decreases in flux produced when an orbiting body crosses the stellar disk (Ricker et al. 2015). TESS Objects of Interest (TOIs) are candidates that require additional scrutiny because an aperture-level transit signal can also be produced by stellar variability, instrumental systematics, or an eclipsing source blended with the target. Ground-based time-series photometry can test whether an event occurs on the expected star, at the expected time, and with an approximately compatible depth and duration. Its interpretation depends on an ephemeris that is current enough to place the event inside the observing window.

TOI-3505.01 is associated with TIC 390988385 at $\alpha = 19:48:10.43$, $\delta = +18:41:56.1$ (ICRS; 297.043476° , 18.698914°). The TIC v8 snapshot gives $T = 10.9372$, an effective temperature of 6220 K, and a stellar radius estimate of $1.335 R_\odot$ (Stassun et al. 2019). The TOI catalog quantities frozen for this analysis (Guerrero et al. 2021) are $P = 2.9151556 \pm 0.0000117$ d, epoch 2459793.534385 BJD_{TDB}, depth 2.91 ppt, and duration 2.004 h. A close companion reported in the project context is separated by 0.517 arcsec with $\Delta I = 1.7$ mag. That separation is far below a TESS pixel and below the effective resolution of the ground apertures used here.

The GMU program obtained an R-band sequence during the night of UT 2022 July 22. The recovered scheduling row placed ingress and egress inside that sequence, yet the reduced light curve did not show a transit. A null observation can have several explanations: inadequate precision, an astrophysical false positive, an event outside the data, or a stale prediction. The public record later acquired additional TESS sectors and ground reports, creating enough timing leverage to distinguish among those possibilities without treating any report-level result as an independent validation.

In this paper, we combine all four public TESS sectors available for TOI-3505.01, construct a 27-event timing solution, quantify its sensitivity to analysis choices, and reconstruct the historical 2022 schedule. To our knowledge, this is the first dedicated study to combine all four publicly available TESS sectors of TOI-3505.01 and trace the 2022 GMU null result to a schedule window consistent with a superseded ephemeris. This deliberately narrow novelty statement does not claim the first detection, classification, follow-up observation, validation, or confirmation of the candidate.

Section 2 inventories the target, TESS products, GMU sequence, and public follow-up context. Section 3 describes the ground and TESS analyses, timing controls, schedule reconstruction, and localization screens. Section 4 reports the measurements. Section 5 interprets the null observation and defines the limits of the candidate-level result. Section 6 summarizes the conclusions and future work.

2. Observations

In Section 2.1 we present TOI-3505.01, its candidate properties, and its host-star properties from the TESS Input Catalog, Gaia, and the public follow-up record. In Section 2.2 we present the four TESS sector light curves and the official validation products. In Section 2.3 we summarize the observational data collected with the George Mason University 0.8 m telescope. In Section 2.4 we present the public ground-based, imaging, and spectroscopic follow-up already on file for this target.

2.1. Catalog and public status

We queried the TESS Input Catalog version 8 and Gaia Data Release 3 around the target (Stassun et al. 2019; Gaia Collaboration et al. 2023). The frozen TIC cone-search result contains a contamination ratio of 0.547. We

use that number only as a warning that the aperture is blended; it is not a correction applied to a final transit depth. The Exoplanet Follow-up Observing Program for TESS (ExoFOP-TESS) target snapshot was retrieved on 2026 August 22, and the dispositions were re-checked against the live ExoFOP and NASA Exoplanet Archive records on 2026 August 29. Both still list TOI-3505.01 as a planet candidate; ExoFOP retains the annotation “Active planet candidate” and lists no confirmed planet for the target. The TOI row was last modified at ExoFOP on 2025 July 31, before either check.

2.2. TESS photometry

TOI-3505.01 has public TESS photometry in Sectors 14, 41, 54, and 81. We used MIT Quick-Look Pipeline (QLP) products in every sector for a like-for-like recovery and depth comparison (Huang et al. 2020a,b). For the primary timing analysis, QLP is the available series in Sectors 14 and 41; the higher-cadence SPOC PDCSAP light curves were selected in Sectors 54 and 81 (Jenkins et al. 2016). We retained finite measurements with finite errors and $\text{QUALITY} = 0$. QLP and SPOC reductions of a given sector are alternative measurements of the same TESS observations, not independent detections.

Table 1. TESS data used in the sector comparison and primary timing fit. Depths are the observed QLP aperture depths from the uniform first-pass box analysis and are not dilution corrected. The pipeline and cadence columns identify the series selected for the final event timing.

Sector	Public light-curve span (UTC)	Timing pipeline	Cadence (min)	Accepted timings	QLP depth (ppt)
14	2019 Jul 18–Aug 14	QLP	30	4	3.73 ± 0.42
41	2021 Jul 24–Aug 20	QLP	10	8	3.02 ± 0.21
54	2022 Jul 09–Aug 04	SPOC	2	9	2.68 ± 0.30
81	2024 Jul 15–Aug 10	SPOC	2	6	3.13 ± 0.24

The TESS products were obtained from the Mikulski Archive for Space Telescopes (MAST). The four quality-filtered QLP series span 2019 July to 2024 August, while the accepted event midpoints provide a 5.05 yr timing baseline. Official SPOC Data Validation reports are available for Sectors 54 and 81 and for a combined search using those same two sectors. We treat those products as same-observation method checks.

TOI-3505.01 in each TESS sector

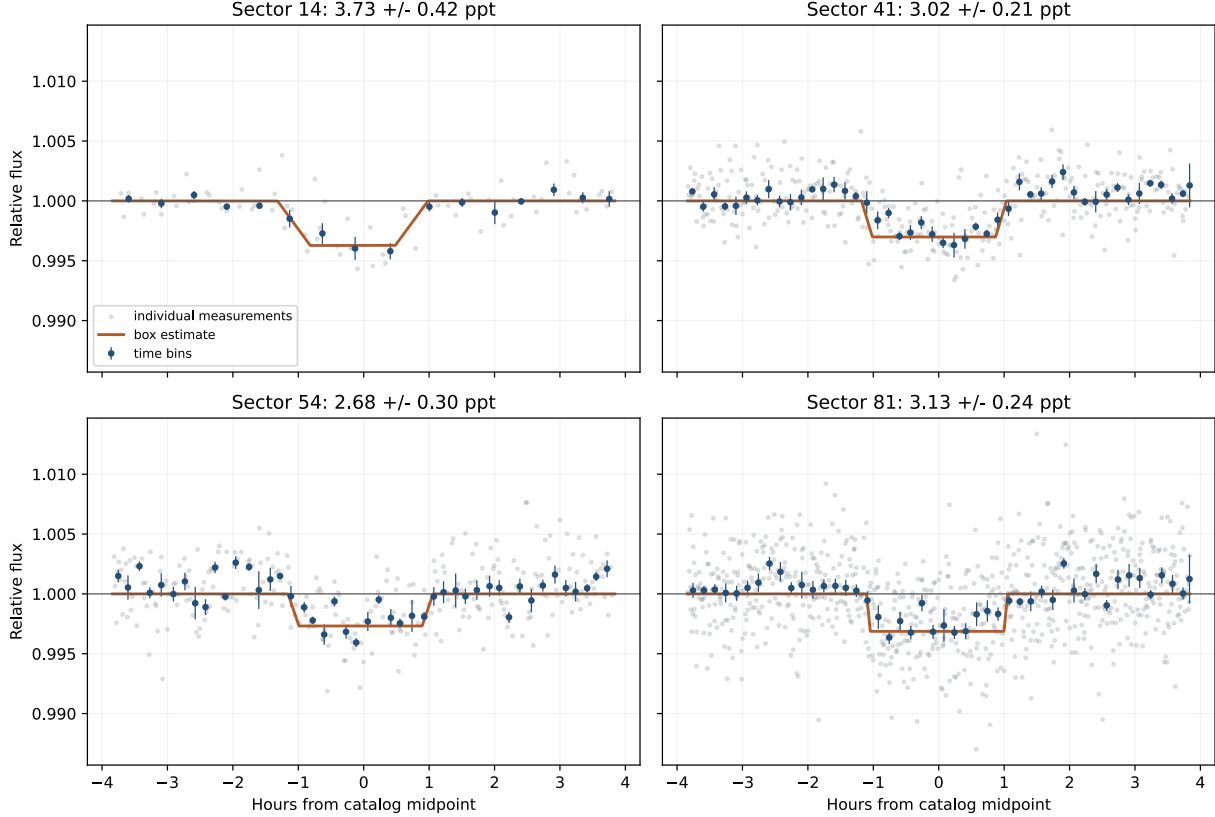


Figure 1. Uniform QLP recovery in all four TESS sectors, folded on the catalog ephemeris. Pale points are individual quality-filtered measurements, blue points are time bins, and the orange line is the exposure-integrated box estimate used for the first-pass sector depth comparison. The sector depths are observed aperture depths, not dilution-corrected physical depths. The final timing analysis replaces this box shape with an exposure-integrated trapezoid and uses SPOC two-minute photometry where available.

2.3. GMU 0.8 m sequence

We observed TOI-3505.01 through an R filter with 50 s exposures. The aligned analysis set contains 281 images from $\text{BJD}_{\text{TDB}} 2459782.598235$ to 2459782.809459 , a duration of 5.069 h. The schedule row gives local-clock ingress at 00:15 and egress at 01:54. The project owner confirmed that these clocks are Eastern time; America/New_York was EDT (UTC-4) on the observing night. The converted window is $\text{BJD}_{\text{TDB}} 2459782.682401$ – 2459782.751151 and is fully contained in the analyzed sequence.

The sequence is shorter than the night the schedule requested. Against a planned 7.75 h range, observing began 1.06 h late and ended 1.62 h early, delivering 65% of the planned window. The schedule row carries the note “bad RA jumps at the beginning of the night”, which is consistent with the late start; no separate observing log survives to confirm it. The delivered sequence nonetheless brackets the scheduled event on both sides, with 2.02 h of pre-ingress and 1.40 h of post-egress baseline, so the fixed-window test in Section 3.1 rests on baseline measured before and after the event rather than on one side alone.

The original workbook, its formulas and revision history, its Transit Info file, and a clock-synchronization record are not available. We preserve those missing records as provenance limitations rather than treating the schedule row as a complete ephemeris.

2.4. Public ground follow-up context

The ExoFOP snapshot contains seven time-series rows representing 6 unique observing nights; two rows duplicate the same GMU observation on UT 2021 June 28. The reports include earlier off-window observations, later

reports describing on-target events, and a tentative MuSCAT2 egress. We did not re-reduce those external light curves. Only the MuSCAT2 report supplies a fitted midpoint uncertainty suitable for the named external timing control.

Table 2. Unique public ExoFOP time-series nights in the frozen 2026 August 22 snapshot. “Context” records report-level information; it does not mean that the underlying data independently validate the candidate.

UT date	Facility	Filter	Coverage	Use here
2021-06-28	GMU 0.8 m	R	Full	Context only
2021-10-15	ULMT 0.6 m	rp	Full	Context only
2022-06-14	KeplerCam 1.2 m	ip	Full	Report-level context
2023-05-05	CMO-SAI-RC-600 0.6 m	g-prime	Full	Report-level context
2023-07-11	OAA 0.4 m	lc	Full	Report-level context
2023-07-14	TCS 1.52 m	g, r, i, z_s	Egress	Timing control only

The same public record also holds high-resolution imaging and spectroscopy that this work does not analyze. SOAR/HRCam speckle imaging reported in 2021 (Tokovinin 2018; Ziegler et al. 2020) resolves the 0.517 arcsec companion at $\Delta I = 1.7$ mag, and Shane/ShARCS adaptive-optics imaging on UT 2021 July 19 reports the same companion at 0.51 arcsec with $\Delta J = 1.53$ and $\Delta K_s = 1.59$ mag, plus five additional sources between 6 and 8 arcsec. Spectroscopic metadata list three TRES epochs in 2021 and two Keck/HIRES entries in 2022; one HIRES note reports six velocities collected between UT 2022 September 1 and 12. The public ExoFOP observing notes report that the three TRES velocities, spanning 2.52 km s^{-1} , are in phase with the photometric ephemeris and are consistent with a companion of approximately 10 Jupiter masses, with the reporting astronomer noting a fair amount of scatter and placing the target on hold pending a multi-order analysis. We list these to establish what constraints already exist for this target. Their analysis is outside the scope of a timing and archival study: no velocity is re-reduced, no orbit is fitted, and no mass is adopted here.

Table 3. Public high-resolution imaging and spectroscopy on file for TOI-3505 in the frozen snapshot. These records are inventoried, not analyzed, in this work.

UT date	Facility	Kind	Result on file	Use here
2021-10-02	SOAR / HRCam	Speckle	0.517 arcsec, delta I = 1.70	Companion parameters
2021-07-19	Shane 3 m / ShARCS	Adaptive Optics	0.51 arcsec, delta J = 1.53, delta Ks = 1.59; five sources between 6 and 8 arcsec	Inventory only
2021	TRES	Spectroscopy	3 epochs on file; velocities in phase with the photometric ephemeris, reported as consistent with about 10 Jupiter masses, on hold	Inventory only
2022	Keck / HIRES	Spectroscopy	2 epochs on file	Inventory only

3. Analysis

In Section 3.1 we present the calibration, registration, and differential aperture photometry applied to the GMU sequence. In Section 3.2 we present the measurement of individual TESS event times and the linear ephemeris fitted to them. In Section 3.3 we present the resampling, selection, pipeline, and model-comparison controls used to test that ephemeris. In Section 3.4 we present the archival reconstruction of the 2022 observing schedule. In Section 3.5 we present the localization and contamination screens, including the nearby-star search. In Section 3.6 we present two further tests bearing on the standard false-positive scenarios.

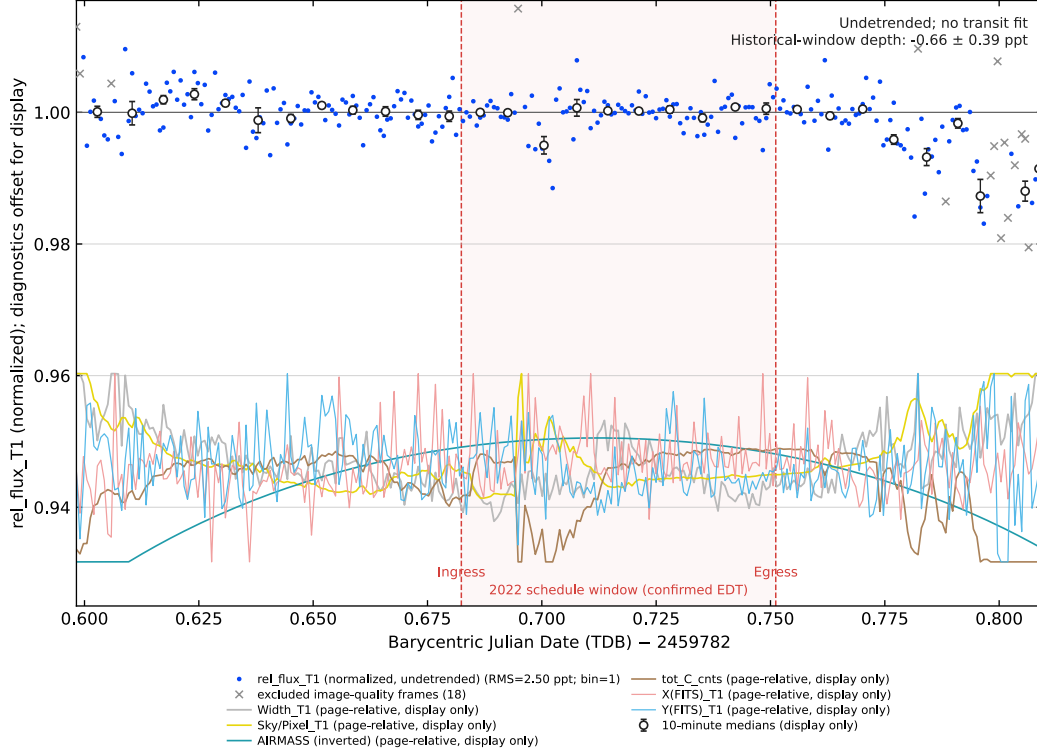
3.1. Ground reduction and differential photometry

We reduced each science image as (science – median 50 s dark) / median normalized R flat. No separate bias subtraction, dark-time scaling, cosmic-ray filter, or interpolating registration was applied. We registered 281 frames to the first image using integer-pixel shifts derived from 25 reference stars. The median post-registration stellar-position RMS was 0.526 pixel.

We used AstroImageJ-style differential aperture photometry (Collins et al. 2017) with a 25 pixel source radius, a 70–139 pixel sky annulus, and the summed counts of ten fixed comparison stars, C2–C11. The final curve uses 263 measurements and excludes 18 measurements for predeclared condition failures or a field-wide image artifact. The seeing profile, field image with apertures, plate-solved reference frame, delta-magnitude versus RMS plot, and nearby-star disposition table corresponding to TFOP SG1 files 5–7 and 9–10 are included in the repository products. Target faintness by itself was never an exclusion criterion. The adopted relative flux was not detrended: none of the tested airmass, sky, width, position, comparison-count, centroid, or combined models passed the blocked cross-validation promotion rules.

We measured the ground precision with the robust scatter of the unbinned adopted flux. For the historical-window test, we fit a straight local baseline plus a fixed, exposure-integrated box between the schedule ingress and egress. The event time and duration were not optimized. We then injected a 2.91 ppt box at the same fixed times and repeated the fit. This same-window injection tests the arithmetic and the sensitivity of this light curve at the historical window; it is not a target-only pixel simulation and does not solve dilution.

Because the schedule window is itself in question, we also searched the sequence independently of it. Holding the duration at a published TESS value and leaving the depth free, we scanned the event midpoint across the night in two-minute steps and refitted the same baseline-plus-box model at each trial position. We required each trial event to be at least 85% sampled and to retain at least twenty baseline measurements on each side of it. That two-sided requirement matters: without it, trial events that hang off the end of the sequence let the baseline slope trade against depth, and the degraded final half hour of the night - where the scatter is roughly four times the mid-run value - produces a large spurious dimming. At every admissible midpoint, we then injected the corresponding published depth into the observed curve and refitted it. We report the fraction of those exact-depth injections whose total fitted depth exceeds a formal 3σ threshold. The weighted-least-squares covariance is locally scaled by reduced χ^2 but does not model temporal correlation, so the recovery fraction is a phase-sampling diagnostic for this residual realization, not a detection probability or ensemble completeness estimate. We quote no Gaussian false-alarm probability and make no global exclusion from the formal limits alone.



Schedule times confirmed as Eastern; the current-catalog transit is outside this sequence.

Figure 2. Final GMU 0.8 m R-band differential light curve. Blue points are the 263 accepted measurements, open circles are ten-minute display medians, and gray crosses are excluded image-quality frames. The red interval is the confirmed-EDT 2022 schedule window. Width, sky, inverted airmass, comparison counts, and detector positions are scaled into the lower display band only; they do not enter the adopted flux or fixed-window depth. The legend is outside the data axes.

3.2. TESS event times and the linear ephemeris

The uniform sector analysis searched a local period and duration grid with a finite-exposure box and one straight baseline per event. For the final timing analysis, we used an exposure-integrated symmetric trapezoid. Its flat-to-total duration fraction, $T_{23}/T_{14} = 0.378$, was derived from the SPOC multi-sector radius ratio and impact parameter. This approximate shape represents the long ingress and egress of the near-grazing solution without claiming a limb-darkened physical fit.

Each event was measured within a ± 0.20 d local window. We accepted a midpoint if the local window was complete, its sampled-window coverage was at least 0.70, the fitted depth signal-to-noise ratio was at least 2.5, and its timing uncertainty was no greater than 0.03 d. These rules retained 27 midpoints: four in Sector 14, eight in Sector 41, nine in Sector 54, and six in Sector 81. QLP was used in Sectors 14 and 41, and SPOC two-minute data were used in Sectors 54 and 81, so each transit entered once.

We fit the accepted midpoints with

$$T(E) = T_0 + P E,$$

where E is the integer cycle relative to the catalog epoch. Weighted least squares used the event-level midpoint errors. When χ^2_{ν} exceeded one, we multiplied the covariance-derived parameter errors by $\sqrt{\chi^2_{\nu}}$; for the adopted fit

the scale is 1.141. We retained the epoch–period covariance when propagating prediction uncertainties. Times are reported in BJD_{TDB} ; conversions followed the barycentric timing conventions of Eastman et al. (2010) and used Astropy (Astropy Collaboration et al. 2022).

3.3. Timing robustness and model comparison

We evaluated whether the period depended on individual events, sector sampling, event-selection thresholds, or model complexity. First, we collapsed each sector to one conservative timing anchor; each anchor uncertainty was the larger of its formal weighted-mean error and its empirical within-sector standard error. Second, we refit after omitting each sector and after omitting each accepted event. Because there are only four sector clusters, the delete-one-sector jackknife is a sensitivity scale rather than a calibrated replacement for the primary uncertainty. Third, we repeated the fit at depth-S/N thresholds of 3 and 4.

We compared the linear ephemeris with a pivoted quadratic timing model,

$$T(E) = a + b(E - E_p) + q(E - E_p)^2,$$

using the Bayesian information criterion (BIC; Schwarz 1978). Finally, we appended the public MuSCAT2 midpoint, $\text{BJD}_{\text{TDB}} 2460140.4350453 \pm 0.000639$, to a separate 28-point control. The report describes a tentative egress, so this point never enters the headline TESS-only result.

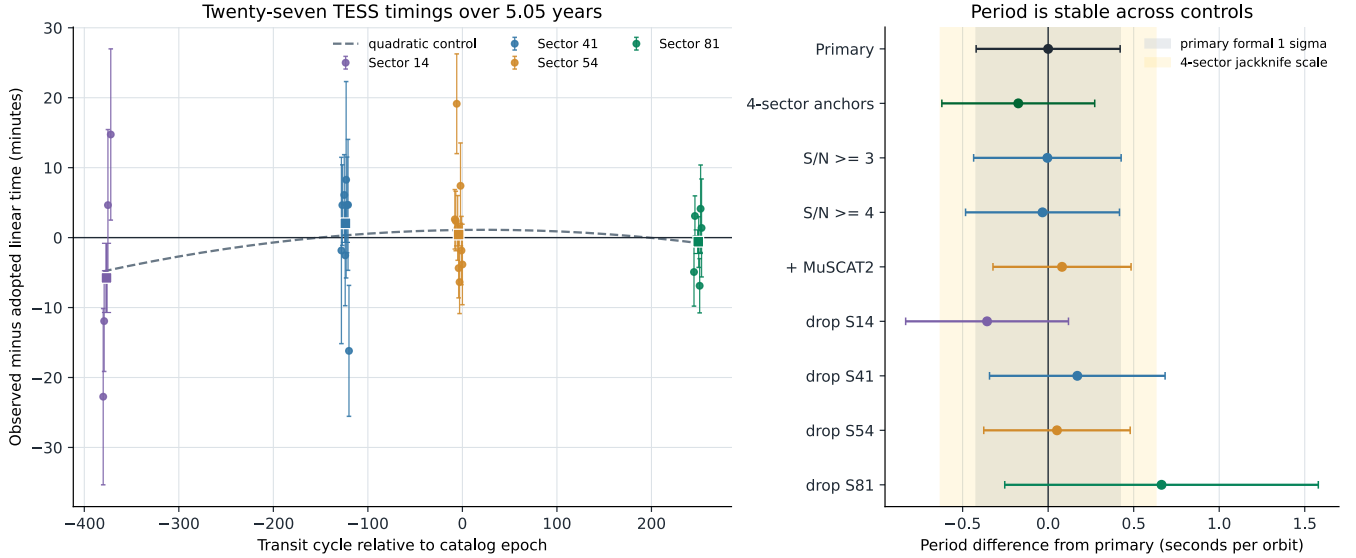


Figure 3. Ephemeris robustness. Left: observed minus adopted-linear timing residuals for 27 accepted TESS events over 5.05 yr; squares are conservative sector anchors and the dashed curve is the quadratic control. Right: period differences from the primary TESS-only fit. The gray band is the primary formal 1σ interval and the gold band is the four-sector jackknife sensitivity scale. QLP and SPOC entries are not independent when derived from the same sector.

3.4. Reconstruction of the 2022 schedule

The schedule row visibly lists 2.9151488 d but does not preserve its prediction epoch. We therefore searched the public report history for a prior event whose propagated ingress, midpoint, and egress could generate the schedule cells. A 2021 October 15 ULMT report gives all three BJD_{TDB} markers. A later CMO report records both a superseded period of 2.9174250 d and the revised 2.9151488 d period, and states that the revised Sector 41 ephemeris moved prior events by approximately 20 h.

We propagated each ULMT marker by the same integer number of cycles,

$$T_{2022, \text{marker}} = T_{2021, \text{marker}} + 96 \times 2.9174250 \text{ d},$$

and compared the result directly with the confirmed-EDT schedule markers. As controls, we propagated the same 2021 midpoint with the revised period and calculated the nearest event from the adopted TESS ephemeris. This is an archival consistency reconstruction. Without the original workbook, agreement cannot distinguish a stale formula from manually copied timing cells.

Table 4. Marker-by-marker schedule reconstruction. Positive residuals mean the schedule marker is later than the propagated marker.

Marker	2021 BJD _{TDB}	+96 cycles	2022 schedule	Schedule – propagation (s)
Ingress	2459502.6093000	2459782.6821000	2459782.6824006	+26.0
Midpoint	2459502.6441000	2459782.7169000	2459782.7167758	-10.7
Egress	2459502.6790000	2459782.7518000	2459782.7511510	-56.1

TOI-3505.01: archival reconstruction of the 2022 schedule

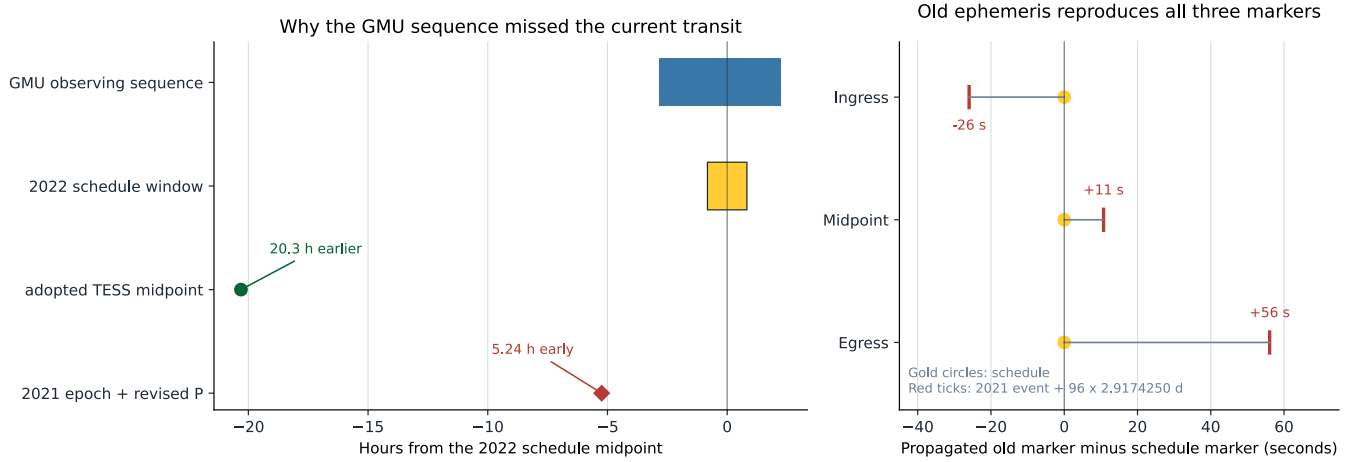


Figure 4. Archival reconstruction of the 2022 schedule. Left: the GMU sequence covered the schedule window, while the adopted TESS event occurred 20.31 h before the schedule midpoint; applying the visible revised period to the 2021 midpoint predicts a time 5.24 h early. Right: propagating each 2021 marker by 96 cycles at 2.9174250 d reproduces ingress, midpoint, and egress within 56.1 s. The match is strong evidence of consistency, not proof of the missing workbook history.

3.5. Localization and contamination screens

We compared circular TESScut apertures from 1.5 to 4.0 pixels and constructed difference images at a fixed 3.0 pixel reference radius. In every sector, the difference centroid lies within one TESS pixel of the target coordinates. This localizes the signal to the target system at TESS resolution but cannot distinguish the primary star from the 0.517 arcsec companion.

For the ground images, we measured all 409 TIC sources within 2.5 arcmin bright enough to mimic a 2.91 ppt event under a total-eclipse screen, applying the -0.5 mag correction to ΔT that the TFOP SG1 guidelines specify to compensate for the difference between the TESS band and the redder follow-up filter. Evaluated at the confirmed schedule window, 288 sources are cleared, 4 are blended with the target aperture, 115 are too faint to yield a decisive limit, and 2 are measured but without a limit deep enough to exclude the required eclipse. Omitting the band correction changes the cleared count by two, so the screen is not sensitive to that choice.

For the 2 measured but uncleared sources we asked whether the apparent dimming has the shape of an eclipse, by comparing the post-egress baseline with the pre-ingress baseline. Neither does. In both the flux fails to return to its pre-ingress level, so the apparent depth is a monotonic trend across the night rather than an event, and in both the night-long scatter exceeds the apparent depth. This is a Python reimplement of the screen rather than the AstroImageJ macro itself, and it does not resolve the close companion; neither do the TESS difference images.

We also calculated dilution scenarios using the TIC contamination ratio and ΔI as a TESS-band proxy. The I band spans roughly 700–900 nm and therefore sits inside the 600–1000 nm TESS bandpass, and the Shane ΔI

brackets it from the red side, so the companion contrast is better constrained than a generic optical-to-TESS extrapolation would be. Under the screening budget the companion contributes a flux ratio of 0.209 relative to the target, leaving the target 0.569 of the light in the aperture; if the 2.91 ppt catalog depth is an uncorrected observed depth, the implied host depth is 5.11 ppt.

We do not adopt that scenario, for a reason worth stating explicitly: SPOC PDCSAP photometry is already corrected for catalog crowding, whereas the QLP series used in Sectors 14 and 41 is not, so Table 1 mixes corrected and uncorrected depths and a further contamination correction applied uniformly would double-count for the SPOC sectors. Resolving that, and the companion’s true TESS-band contrast, is a prerequisite for a physical depth. We therefore report no dilution-corrected depth and no planet radius from this work.

3.6. False-positive scenario tests

The public archive supports two further tests that bear directly on the standard false-positive scenarios, and we apply both to values as published rather than re-reducing them.

First, a blended eclipsing binary generally produces a transit depth that varies with wavelength, because the blended source and the target differ in colour. The MuSCAT2 report gives simultaneous depths in g , r , i , and z_s (Narita et al. 2019). We regressed those depths against the effective wavelength of each band. The report publishes no per-band uncertainty, so we use the scatter about the mean only to describe the scale of the variation and the fitted slope. It does not support a calibrated achromaticity test.

Second, an eclipsing stellar companion on the target would move the host by a velocity semi-amplitude far larger than a planetary one. We computed the circular, edge-on semi-amplitude that companions of 0.1, 0.3, and 0.6 $M_\odot$ would produce at the adopted period, assuming a 1.25 $M_\odot$ host, and compared them with the span of the three public reconnaissance velocities. This is a scale comparison, not an orbit fit: the observed quantity is the range of three recon velocities, not a fitted amplitude, and the host mass is assumed rather than measured.

4. Results

In Section 4.1 we report the ground light curve, the fixed-window measurement, and the whole-sequence transit search. In Section 4.2 we report the four-sector signal and the adopted ephemeris. In Section 4.3 we report the robustness controls. In Section 4.4 we report the reconstruction of the historical schedule. In Section 4.5 we report the false-positive scenario tests.

4.1. Ground light curve

The 263 accepted ground measurements have an unbinned robust scatter of 2.497 ppt. The fixed historical-window fit gives -0.658 ± 0.395 ppt, or -1.67σ when positive depth denotes dimming. The negative value is therefore not a transit-like event. The adopted current-catalog prediction was already 17.39 h before the first accepted ground exposure, so the ground sequence did not observe the current predicted event.

Adding a 2.91 ppt same-window box recovered an incremental depth of 2.910 ppt. Including the measured negative control at that location, the total recovered dimming has signal-to-noise 5.71. The injection demonstrates sensitivity to a catalog-scale dimming at the fixed historical window; it does not establish sensitivity to every transit shape or blend scenario.

The whole-sequence search finds no compelling event at any time it can test. Requiring a well-sampled event with baseline on both sides restricts the catalog-duration midpoints to 1.44–3.47 h after the first exposure, because the sequence is only about twice the transit duration. Within that range the largest fitted dimming is 0.93 ± 0.42 ppt, a formal single-midpoint S/N of 2.20. The median formal 3σ WLS upper bounds are 0.99 ppt for the 2.00 h catalog duration and 0.68 ppt for the 2.71 h SPOC duration, against published depths of 2.91 and 3.29 ppt. These bounds omit temporal covariance and are therefore not interpreted as global exclusions. When the corresponding published depth is injected into the observed curve, the fitted event exceeds the formal 3σ threshold at 50/62 cat-

alog-duration midpoints (80.6%; minimum formal S/N 0.68) and 40/42 SPOC-duration midpoints (95.2%; minimum formal S/N 2.84). Neither recovery fraction is 100%, so the sequence does not exclude a published-depth event at every admissible phase (Figure 5).

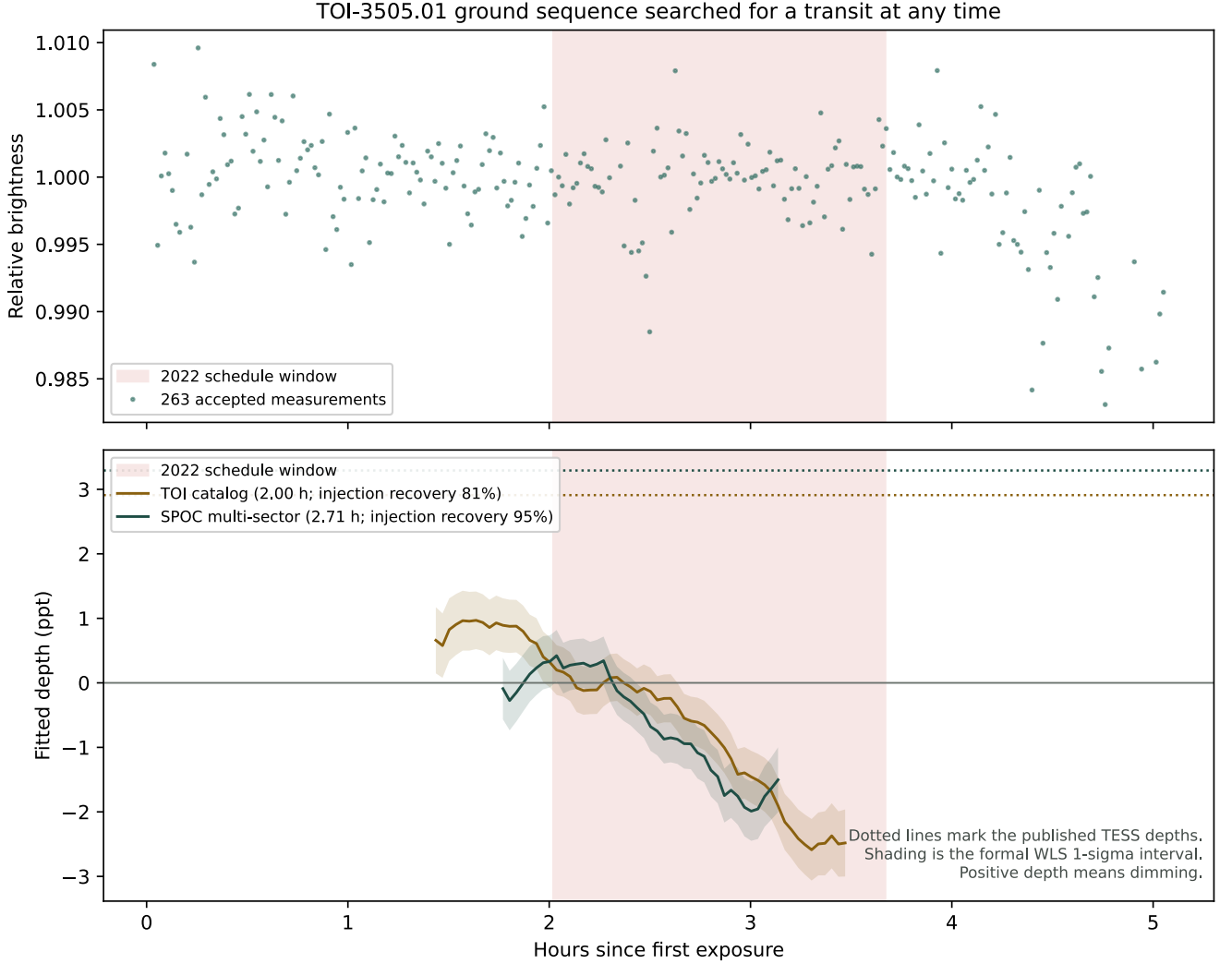


Figure 5. Whole-sequence transit search. Top: the 263 accepted measurements, with the 2022 schedule window shaded. Bottom: fitted depth against trial midpoint for the catalog and SPOC durations; shading gives formal WLS 1σ intervals and dotted lines mark the published TESS depths. Legend percentages give the fraction of exact-depth injections above a formal 3σ threshold. They describe phase sensitivity in this observed residual realization, not a detection probability. Positive depth denotes dimming. The two-sided-baseline requirement excludes the ends of the night.

4.2. Four-sector signal and adopted ephemeris

The uniform QLP analysis recovers a recurring dip in all four sectors with observed depths from 2.68 to 3.73 ppt (Figure 1). These values are consistent at the level expected for different cadences and apertures, but they are not physical, dilution-corrected depths. The official SPOC reports for Sectors 54 and 81 recover the same signal and do not show a strong odd/even, secondary-event, or centroid alert. Because they reuse the same TESS pixels, those reports are diagnostic comparisons rather than independent evidence.

The adopted linear fit is

$$T_0 = 2459793.53115 \pm 0.00088 \text{ BJD}_{\text{TDB}}, \quad P = 2.9151516 \pm 0.0000049 \text{ d}.$$

It has $\chi^2_{\nu} = 1.301$ after the stated uncertainty scaling and an RMS timing residual of 8.51 min. Its formal period uncertainty is 2.40 times smaller than the frozen TOI catalog uncertainty and 1.43 times smaller than the SPOC

multi-sector value. Precision comparisons do not make the fit independent of those source observations.

Table 5. Linear ephemeris estimates. The primary row is the only headline result. The four-sector and MuSCAT2 rows are controls with the limitations described in Section 3.

Fit	N	Period (d)	1 σ error (d)	$\chi^2\nu$	Role
Primary TESS linear	27	2.9151515671	0.0000048720	1.30	Adopted
Four sector anchors	4	2.9151495461	0.0000051714	0.49	Conservative aggregation
QLP only	24	2.9151569718	0.0000062841	1.10	Pipeline control
TESS + MuSCAT2	28	2.9151525056	0.0000046714	1.28	External control only

4.3. Robustness controls

The four conservative sector anchors give $P = 2.9151495 \pm 0.0000052$ d, consistent with the primary result. The delete-one-sector jackknife scale is 0.0000073 d, or 1.50 times the primary formal error. We do not substitute this four-cluster value for the event-level uncertainty. The delete-one-event jackknife scale is 0.0000058 d, and no single-event deletion shifts the period by more than 0.0000032 d. Raising the minimum depth S/N to 3 or 4 leaves the period within the primary formal uncertainty.

The quadratic coefficient is $q = -2.544\text{e-}08 \pm 2.295\text{e-}08$ d cycle⁻² (-1.11 σ). It improves χ^2 by only 1.58 while adding a parameter, yielding $\Delta\text{BIC} = \text{BIC}_{\text{quadratic}} - \text{BIC}_{\text{linear}} = +1.71$. The timing data therefore do not favor curvature. Adding the tentative MuSCAT2 timing only as an external control gives 2.9151525 ± 0.0000047 d, a 4.12% formal precision change, and remains consistent with the TESS-only period.

4.4. Historical schedule

The adopted ephemeris places the relevant midpoint 20.31 h before the schedule midpoint. Propagating the public 2021 markers by 96 cycles with the superseded 2.9174250 d period instead reproduces the 2022 schedule: the ingress, midpoint, and egress residuals have maximum absolute magnitude 56.1 s (Table 4). Applying the visible 2.9151488 d value to the same 2021 midpoint predicts an event 5.24 h before the schedule midpoint. The combined evidence is strongly consistent with timing cells retained from the superseded ephemeris while the displayed period cell was updated.

4.5. False-positive scenario tests

The four MuSCAT2 depths average 2.43 ppt with a scatter of 0.30 ppt and are not monotonic in wavelength. A descriptive regression against effective wavelength gives a slope of -0.032 ppt per 100 nm and a residual-based slope scale of 0.124 ppt per 100 nm. Because the report supplies no per-band measurement uncertainties, their ratio is not a calibrated significance and supports only the qualitative statement that no monotonic chromatic trend is apparent across 477–870 nm (Figure 6, left).

An eclipsing stellar companion at the adopted period would produce a velocity semi-amplitude of at least 12 km s⁻¹ for the smallest case considered, 5 times the 2.52 km s⁻¹ span of the public reconnaissance velocities (Figure 6, right). The SPOC reports add that neither sector shows a significant odd/even depth difference, that the weak-secondary multiple-event statistics remain below detection, and that neither sector is flagged as a suspected eclipsing binary (Twicken et al. 2018; Li et al. 2019).

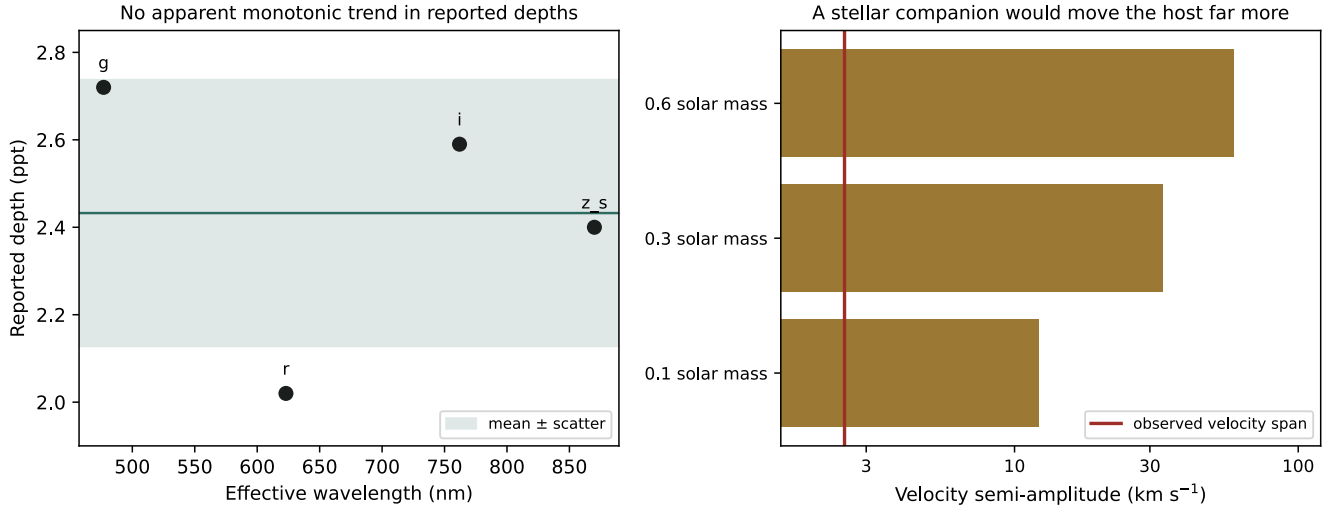


Figure 6. False-positive scenario tests. Left: the four reported MuSCAT2 depths against effective wavelength, with the mean and residual scatter shaded. The absence of published per-band uncertainties makes this comparison descriptive. Right: circular, edge-on velocity semi-amplitudes for stellar companions at the adopted period, against the span of the three public reconnaissance velocities. Both panels use published values and are not re-reductions.

5. Discussion

In Section 5.1 we interpret the 2022 null observation. In Section 5.2 we discuss the precision and scope of the timing solution. In Section 5.3 we place the candidate in the context of the transiting-planet population. In Section 5.4 we work through the standard false-positive scenarios. In Section 5.5 we set out what the candidate status and the unresolved blends do and do not permit us to conclude. In Section 5.6 we define the literature and novelty boundary of this work.

5.1. Interpretation of the 2022 null

The ground light curve alone answers a limited question: no transit-like dimming is measured inside the exact window the 2022 schedule told the observers to cover. The same-window injection shows that a catalog-scale box would have been recovered in that portion of the data. The updated TESS ephemeris then resolves the apparent conflict by placing the event predicted by the adopted linear ephemeris outside the sequence.

The schedule reconstruction provides a specific historical explanation. The three marker cells are jointly reproduced to better than one minute by a public 2021 prediction propagated with the superseded period. That residual is close to the floor the source data can express: the schedule records its ingress and egress only to the nearest minute, so agreement much better than the 56.1 s achieved here could not be demonstrated from these cells even if the underlying formula were recovered exactly. A coincidental match of ingress, midpoint, and egress is less plausible than a single midpoint coincidence because it also preserves the event duration. Nevertheless, we cannot inspect the missing workbook formulas or revision history. We therefore describe the result as a strongly supported archival reconstruction, not proof that a particular spreadsheet formula caused the error.

The null observation should not be used as evidence that the TESS signal disappeared. It is an off-transit observation under the adopted ephemeris. Conversely, explaining the schedule does not turn the ground data into a transit detection.

5.2. Timing precision and model scope

The long, four-sector baseline supplies most of the period leverage. Sectors 14 and 81 are separated by more than five years, while the two-minute SPOC sampling in the later sectors improves individual midpoint precision. The consistency of the period under conservative sector aggregation, stricter selection, event deletion, sector deletion,

QLP-only processing, and the external MuSCAT2 control indicates that no tested event or threshold alone determines the headline value.

The formal uncertainty remains conditional on the approximate trapezoid, the event-level timing errors, and the single linear ephemeris. The four-sector jackknife is larger but is based on only four clusters. Reporting both quantities distinguishes a fit uncertainty from a coarse sensitivity scale. The positive ΔBIC for the quadratic model means that these timings do not justify an added curvature parameter; it does not prove that the physical period is exactly constant.

5.3. The candidate in context

The published parameters place TOI-3505.01 among short-period giant-planet candidates. The TOI catalog lists a planet radius of $7.46 R_{\oplus}$ and an insolation of $505 S_{\oplus}$, and the host is a late-F dwarf with $T_{\text{eff}} = 6220 \text{ K}$ and $1.335 R_{\odot}$. On a 2.9151516 d orbit, the catalog radius is larger than Neptune’s but below Jupiter’s. We quote these as catalog values: this work does not fit a radius, and the standing dilution limits described in Section 5.5 are the reason.

The geometry is the complicating factor. The SPOC multi-sector solution gives an impact parameter of 0.92, which is close to grazing. Near-grazing transits couple depth, duration, and impact parameter tightly, so a depth measured without a documented dilution model does not translate cleanly into a radius. This is why we replace the box with an exposure-integrated trapezoid for timing (Section 3.2) but stop short of a physical fit, and it is the main reason a radius from this work would be less reliable than the catalog value we cite.

5.4. False-positive scenarios

Taking the standard scenarios in turn, each is constrained by at least one line of evidence, and none is excluded outright.

A nearby eclipsing binary is disfavoured by the screen in Section 3.5: of 409 catalogued neighbours bright enough to matter, none that could be measured shows an eclipse-shaped event of the required depth, and the two that the screen cannot clear on depth alone are monotonic trends across the night rather than events. Public nearby-star checks accompanying the KeplerCam, ULMT, and MuSCAT2 observations report the same, and the SPOC difference-image centroid sits under $10''$ from the catalogued position in both sectors, so the source of the dimming coincides with the target at TESS resolution. What this leaves open is the population too faint for a decisive limit, and anything inside the target aperture.

A blended eclipsing binary is weakly disfavoured by the lack of an apparent monotonic trend in the four reported MuSCAT2 depths. Without per-band uncertainties, however, those points do not establish calibrated achromaticity. A blend whose colour resembles the target could also evade this screen, so the result is qualitative and does not exclude the scenario.

An eclipsing stellar companion on the target itself is the most strongly disfavoured. It would move the host by at least 12 km s^{-1} , and the public velocities span 2.52 km s^{-1} . The SPOC odd/even, secondary-eclipse, and eclipsing-binary diagnostics agree. The caveat is that a definitive mass awaits the multi-order analysis the public notes leave open.

None of this reaches the scenario that matters most. The $0.517''$ companion is unresolved in every dataset used here, and the tests above constrain the star the spectrograph and the imagers resolved, not the pair. That is why the sections below claim no validation and no radius.

5.5. Candidate status and unresolved blends

The recurring signal and TESS-scale centroid agreement are compatible with a transit on the target system. The public reconnaissance velocities bear on this from a different instrument and a different dataset: the observing note interprets a velocity offset in phase with the photometric ephemeris as consistent with an approximately 10 -Jupiter-mass companion. That report is not a closed result, since its author recorded scatter and left the target on

hold, and we do not re-derive it. Even taken at face value it does not determine which component of the unresolved system is dimming, because a 0.517 arcsec companion is inseparable in the TESS images, in the 25 pixel ground aperture, and in a fiber-fed reconnaissance spectrum. The conditional nearby-star screen also leaves 4 apertures overlapping the target and is not a formal nearby-eclipsing-binary clearance.

These limitations are why we do not apply the scenario dilution correction, infer a planet-to-star radius ratio, calculate a planet radius, or report a false-positive probability. The live public status at the analysis date remains PC. Statistical validation would require a documented combination of resolved imaging, spectroscopy or radial velocities as appropriate, fully characterized light curves, and a formal false-positive analysis. As Section 2.4 records, resolved imaging and spectroscopic epochs for this target already exist in the public archive; what is missing is not the observations but a joint analysis of them, which is beyond the scope of this timing and archival study.

5.6. Literature and novelty boundary

TOI-3505.01 appears in the population-level Ariel target list of Edwards & Tinetti (2022), and public ground follow-up predates this analysis. Exact searches for “TOI-3505.01,” “TOI 3505,” and TIC 390988385 found no dedicated paper or preprint in the literature snapshot, refreshed on 2026 August 29 across arXiv, OpenAlex, the NASA Exoplanet Archive bibliography, and general scholarly web results. The supported novelty is therefore the combination of all four public TESS sectors with the archival schedule reconstruction, not discovery or first follow-up. We additionally confirmed within the observing program that no parallel study of this target is in progress. Because a search can miss an unexpected identifier or unindexed venue, and because that confirmation cannot extend to unpublished work elsewhere, the phrase “To our knowledge” must remain, and the search should be repeated immediately before submission or preprint posting.

6. Conclusions and future work

We analyzed the GMU 0.8 m R-band sequence and all four public TESS sectors of TOI-3505.01. The ground data show no dimming in the confirmed-EDT 2022 schedule window: -0.658 ± 0.395 ppt. The whole-sequence scan contains no formal 3 σ event, but exact published-depth injections cross that threshold at 80.6% and 95.2% of the admissible midpoints for the two tested durations; the data therefore do not support exclusion at every phase. The primary TESS-only fit gives $P = 2.9151516 \pm 0.0000049$ d from 27 events over 5.05 yr. The larger 0.0000073 d delete-one-sector scale records the sensitivity to the four widely separated observing clusters, and the remaining tested controls do not motivate a different headline period. The event predicted by the adopted ephemeris was 20.31 h before the schedule midpoint, while a superseded 2.9174250 d public ephemeris reproduces all three schedule markers within 56.1 s. The simplest supported interpretation is that the 2022 observations followed a stale event window.

Future work should obtain a new, full transit using the updated prediction and an independently verified observatory clock, and should complete the multi-order velocity analysis that the public notes leave open and combine it with the resolved imaging already held in the archive. Those, together with a documented crowding treatment, are what a formal false-positive assessment requires. A joint, exposure-integrated, limb-darkened transit model should be attempted only after the dilution model is documented, and a longer sequence than the one analyzed here would be needed to search for events near the ends of a night. Recovering the original 2022 workbook, formula history, Transit Info file, and clock record would further strengthen the archival interpretation. Until those steps are complete, TOI-3505.01 remains a planet candidate and no dilution-corrected radius should be quoted.

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Data and code availability

The analysis code, frozen catalog and public-report snapshots, derived tables, figures, claim-evidence ledger, decision log, software record, and SHA-256 file manifest are available in the project repository at <https://github.com/mason-cao/nasa-tess-exoplanet-analysis>. TESS mission products were obtained from <https://archive.stsci.edu/missions-and-data/tess>; QLP products are identified by dataset DOI 10.17909/t9-ro86-e880. The public follow-up source is <https://exofop.ipac.caltech.edu/tess/target.php?id=390988385>. The repository preserves exact product filenames and retrieval metadata. The external reports are contextual; their image-level data were not re-reduced in this work.

Software

The reproducible workflow uses Python, NumPy, pandas, SciPy, Matplotlib, Astropy, astroquery, photutils, Lightcurve, and AstroImageJ. Exact installed versions and the source revision are recorded in `outputs/toi3505_research_record/software_versions.json`. Because writing that record is itself a change to the repository, the commit it names is the parent of the commit that carries it. All displayed manuscript values are injected from canonical machine-readable outputs by `src/build_toi3505_paper.py`; the paper build also writes an auditable `manuscript_values.json` record.

References

- Astropy Collaboration et al. 2022, “The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package,” *ApJ*, 935, 167, doi:10.3847/1538-4357/ac7c74.
- Collins, K. A., Kielkopf, J. F., Stassun, K. G., & Hessman, F. V. 2017, “AstroImageJ: Image Processing and Photometric Extraction for Ultra-precise Astronomical Light Curves,” *AJ*, 153, 77, doi:10.3847/1538-3881/153/2/77.
- Eastman, J., Siverd, R., & Gaudi, B. S. 2010, “Achieving Better Than 1 Minute Accuracy in the Heliocentric and Barycentric Julian Dates,” *PASP*, 122, 935, doi:10.1086/655938.
- Edwards, B., & Tinetti, G. 2022, “The Ariel Target List: The Impact of TESS and the Potential for Characterizing Multiple Planets within a System,” *AJ*, 164, 15, doi:10.3847/1538-3881/ac6bf9.
- Gaia Collaboration et al. 2023, “Gaia Data Release 3: Summary of the Content and Survey Properties,” *A&A*, 674, A1, doi:10.1051/0004-6361/202243940.
- Guerrero, N. M., Seager, S., Huang, C. X., et al. 2021, “The TESS Objects of Interest Catalog from the TESS Prime Mission,” *ApJS*, 254, 39, doi:10.3847/1538-4365/abefe1.
- Huang, C. X., Vanderburg, A., Pál, A., et al. 2020a, “Photometry of 10 Million Stars from the First Two Years of TESS Full Frame Images: Part I,” *RNAAS*, 4, 204, doi:10.3847/2515-5172/abca2e.
- Huang, C. X., Vanderburg, A., Pál, A., et al. 2020b, “Photometry of 10 Million Stars from the First Two Years of TESS Full Frame Images: Part II,” *RNAAS*, 4, 206, doi:10.3847/2515-5172/abca2d.
- Jenkins, J. M., Twicken, J. D., McCauliff, S., et al. 2016, “The TESS Science Processing Operations Center,” *Proc. SPIE*, 9913, 99133E, doi:10.1117/12.2233418.
- Li, J., Tenenbaum, P., Twicken, J. D., et al. 2019, “Kepler Data Validation II—Transit Model Fitting and Multiple-planet Search,” *PASP*, 131, 024506, doi:10.1088/1538-3873/aaf44d.
- Narita, N., Fukui, A., Kusakabe, N., et al. 2019, “MuSCAT2: four-color simultaneous camera for the 1.52-m Telescopio Carlos Sánchez,” *JATIS*, 5, 015001, doi:10.1117/1.JATIS.5.1.015001.
- Ricker, G. R., Winn, J. N., Vanderspek, R., et al. 2015, “Transiting Exoplanet Survey Satellite,” *JATIS*, 1, 014003, doi:10.1117/1.JATIS.1.1.014003.
- Schwarz, G. 1978, “Estimating the Dimension of a Model,” *Annals of Statistics*, 6, 461, doi:10.1214/aos/1176344136.
- Stassun, K. G., Oelkers, R. J., Paegert, M., et al. 2019, “The Revised TESS Input Catalog and Candidate Target List,” *AJ*, 158, 138, doi:10.3847/1538-3881/ab3467.
- Tokovinin, A. 2018, “Ten Years of Speckle Interferometry at SOAR,” *PASP*, 130, 035002, doi:10.1088/1538-3873/aaa7d9.
- Twicken, J. D., Catanzarite, J. H., Clarke, B. D., et al. 2018, “Kepler Data Validation I—Architecture, Diagnostic Tests, and Data Products for Vetting Transiting Planet Candidates,” *PASP*, 130, 064502, doi:10.1088/1538-3873/aab694.
- Ziegler, C., Tokovinin, A., Briceño, C., et al. 2020, “SOAR TESS Survey. I. Sculpting of TESS Planetary Systems by Stellar Companions,” *AJ*, 159, 19, doi:10.3847/1538-3881/ab55e9.